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Millets Value Chain Platform
A Digital Marketplace for "Shree Anna"

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I. ABSTRACT

The nutritional value of the millets has provided them with a renewed significance in the world, along with their climate and environmental sustainability in agriculture [12], [13]. Nevertheless, the current millet value chain in India and other developing economies is highly disintegrated, and it is also defined by limited market accessibility, absence of price transparency, lack of supply chain coordination, and low digital integration as the recent studies on value chain dynamics, perception of consumers, and governance gaps within agri-food systems [1], [3], [5], [9]. Low bargaining power, insufficient certification mechanisms, and relying on intermediaries, in turn, hinder the potential of tribal and smallholder farmers to make an income and enter the market [1], [9], [15].

The proposed paper is a concept of a digital Millet Supply Chain Platform that will reduce all financial limits between farmers, startup processors, customers and administrators into a single online platform. The platform facilitates the process of direct-to-consumer and direct-to-startup transactions between farmers and consumers, online seller verification, raw material purchase, selling of processed goods, traceability, price transparency, an ACM (Accessories, Craft & More) waste-to-value system, and online order and payment tracking. Based on the international trends of digitalizing the agri-food system and traceability (Jayalath et al., 2021; Charlebois et al., 2021; Ahmed et al., 2021), the system can rely on the principles of a modular architecture, role-based dashboards, digital monitoring, and sustainability-focused value chain [3], [4], [5], [16], [17].

This strategy aims at integrating value chain optimization models [6], millet-based market knowledge [10], [11], and consumer-centred product strategies [7] to develop a technology-enabled supply chain. The result is a platform which improves the income of farmers, startup development, quality assurance, market transparency, traceability and sustainable millet-based value chains. The solution that has been proposed will eventually create an inclusive, effective, and digitally empowered agricultural market that is congruent with the future demands of the millet economy [13], [17], [19].

Keywords: Millet value chain, digital supply chain, agri-food traceability, online marketplace, sustainable agriculture, farmer-to-consumer platform.

A. Introduction

B. Background and Motivation

Currently, millets have been re-introduced to the world because of their dense nutrient content, low glycemic index, drought-resistance, and the capacity to grow in arid zones [12], [13]. According to the literature compiled in India, Sri Lanka, and in other markets all over the world, millets are very instrumental in enhancing food security, sustaining sustainable agriculture and livelihoods of smallholder farmers and tribal farmers [1], [2], [11]. The demand of the products based on millet, both raw and value-added, keeps growing in the market with the growing consumer awareness and the growing need of healthy foods.

In spite of this potential, the millet market in India is limited in a number of ways. It has been shown that farmers have low profitability due to value chain inefficiencies, absence of storage and processing facilities, intermediate dominance, and insufficient institutional support [9], [8]. The consumer perception research also highlights ineffective availability of products and low product visibility of processed millet products [7]. Moreover, the supply chain operations in developing economies remain not digitalized, which leads to the lack of traceability, disjointed market connections, and inadequate transparency [3], [5].

With these challenges in mind, an urgent need to have a digital platform that will bring together the stake holders, increase their visibility, provide them with traceability and give direct access to the market is obvious. As OECD and developing countries focus on digital governance and agri-food traceability, a technology-based solution will assist the millet farmers, organic startups, and consumers in the process of sustainable value chain development [4], [17], [19].

C. Problem Definition and Objectives

The structural limitations of the current millet ecosystem include:

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- Absence of price and market transparency that gives farmers little bargaining power [1], [9], [15].
- Disjointed supply chain networks that do not allow effective coordination between farmers, processors and buyers [6].
- Excess dependency on middlemen, which only lowers the income earned by the farmers and raises consumer prices [1], [6].
- Poor traceability and quality control particularly in the developing economies where there is very little digital systems [5], [16], [17].
- Weak platforms of selling agricultural by-products, which can also add to the benefits of the circular economy [13], [17].
- Start-up processors who are not able to get good quality raw millets at a constant price and quantity [11], [18].

In accordance with these gaps, the following objectives are presented in the proposed system:

- To create a single digital marketplace that will unite farmers, startup owners and consumers [15], [18].
- To have clear pricing, authenticated sellers and verifiable supply chain [4], [5].
- To assist both the raw and processed millet products market, create more opportunities in value addition [7], [11].
- To offer modules of order management, tracking payments, certifications check, and digital feedback [3], [17].
- To support sustainability aspects like ACM (Accessories, Craft & More), to allow the use of waste as a resource [13], [19].
- To develop a platform that is inclusive and increases the income of farmers, the development of startups and consumer confidence [6], [15].

D. Overview of the Proposed Project.

The project proposed is a Millet Supply Chain Platform, which is a multi-functional digital system, which will include:

- Farmers: Post verified profiles, sell raw agricultural and by-product items, order taking, receive payment and chat with customers [6], [15].
- Entrepreneurs: Purchase raw materials in the farms, advertise the organic processed millet-based food and cosmetics, and monitor the order/payment procedures [11], [18].
- Customers: Purchase the products, watch the nutritional benefits, discuss and exchange recipes [7], [12].
- Admin: Evaluating Registrations, certifications, monitoring functions, feedbacks, and quality control of platforms [4], [17].

The system has modules of:

- Online platform to sell raw and processed millet products.
- Traceability and certification check-up.
- Payment integration (UPI)
- ACM module of using agricultural waste.
- Recipe and interaction with the community.
- Order and delivery tracking
- Feedback and rating systems

All in all, the system addresses critical gaps in the currently existing value chains by empowering farmers, rural startups, and providing quality products to the consumers [13], [17], [19].

E. Outline of the Paper

The rest of this paper is structured in the following way:

- Section 3 is a broad literature review that dwells on millet value chains, digital agricultural supply chains, traceability frameworks, and research gaps found in the current systems.

- Section 4 presents the methodology used in the proposed platform, which contains requirement analysis, system design strategy, development workflow, and testing process.

- Section 5 outlines the system design and implementation, which offers the architectural framework, functional modules, data flow mechanisms, and implementation strategies.

- Section 6 deals with the experimental findings and system performance with focus on usability, system efficiency and better performances in comparison to the traditional supply chain practices.

- Section 7 will deal with the social, ethical and sustainability issues of the proposed digital platform.

- Section 8 is the conclusion of the paper summarizing the significant contributions and research findings.

- Section 9 contains references.

II. Literature Review

A. Traditional Millet Value Chain Issues

Some other reports indicate that the chain of values that support the growing and selling of the crops is very weak and underdeveloped yet the millets are very nutritional and they are resistant of the climate. The research carried out in the tribal region of Attappadi, Kerala estimates that local middlemen are still necessary to the tribal farmers and they limit their bargaining power, resulting in low prices realisation [1].

These concerns are also present in the regional study of Karnataka whereby smallholder millet farmers are facing the challenge of fragmented markets, inefficient aggregation mechanisms and minimal access to organised buyers [7]. Possible motivators to the improvement of the millet chain include Farmer Producer Organizations (FPOs), though they have restricted reach and are not consistent [8].

The former reviews also indicate that even with the growing awareness of millets, the value-added millet products still have to deal with the issue of visibility, familiarity of consumers, and stable supply [9]. Overall, the disadvantages of the old millet value chain are poor market connectivity, physical market dependence with no incentives to encourage farmers to produce more.

B. Digital Supply Chain and Traceability Issues

As the agri-food supply chains within the global context have become digitalized, several countries have demonstrated the way in which digital traceability and channelized channels can enhance transparency and trust. As per the comparative analysis of the OECD countries, the digital traceability systems can help to enhance the food safety, monitor supply chain processes, and improve the governance systems [4].

In non-developed countries, the digitalization is not yet at its dawn. The papers reveal that the transformation of the traditional type of agriculture to the digital one can significantly improve the coordination, minimizing losses, and sustainable governance [3].

A general literature review of the agri-food traceability in the literature reveals that digital ledgers, cloud, and integrated databases are becoming more and more prominent in terms of tracking the origin, price change, and certifications, and tracking the movement of the products along the value chains [5]. The general feeling of these results is that digital solutions offer a profitable path of facilitating millet supply chains, which are at present underdeveloped in regards to the friendliness of their traceability systems.

C. Role of Technology in Agriculture Market

Technology application has taken the overpowering role when it comes to countering the unbearable inefficiencies in the agricultural markets. The studies of the value chain of developing nations attach importance on the importance of the digital relations within the market, formal procurement processes, and definite price cues in improving the welfare of the farmers (Trienekens, 2011).

The studies specifically pertaining to Millet also reveal the growing demand of the digital platforms which enable the sale of the raw Millet and the market of the value-added products which is also growing. Small studies on milllets require the concepts that small processors and start-ups have a hard time with the consistent nature of raw materials and the good market data (Adekunle et al., 2018).

Other articles about value chain development also note that the digital tools have the potential of spurring entrepreneurship in the millet-based products and enable the rural communities to change their path to more high-end processing to add more value to their products (Shanmuka & Shireesha, 2018). As the world is changing its preferences to foods that are healthy and nutritious, the digital farming systems play a great role in mediating between farmers, the processors, and the consumers that use the healthy foods [13].

D. Gaps in Existing System

There are certain gaps which reappear in all the works:

- Lack of unified platforms amongst farmers, processors, consumers and administrators [10], [12].
- Minimal traceability, such that consumers will not be able to easily determine the source of the products, the condition of the farmers, or the certifications [4], [5].
- Inability to see the value-added millet products even when interest of consumers rose [9].
- Limited or insignificant digital support of FPOs, small millet enterprises or rural entrepreneurs [8], [11].
- Lack of integrated mechanisms to harvest by-products or waste-to-value programmes despite the fact that agricultural waste is a promising provider of crafts and organic products [13].
- There is inconsistency in the pricing information and this exposes the farmers to market volatilities [1], [7].

These loopholes indicate that there was a significant gap in terms of an online, participative, multi-stakeholder platform so that it specifically targeted to the millet ecosystem.

E. Summary of Findings

The above literature illustrates that although millets are nutritionally rich, its demand both in the market and in terms of culture, its value chain is not fully efficient at all, production and aggregation, processing and consumption [1], [6], [9]. Traditional supply chains do not imply transparency, coordination and digitalization. At the same time, the papers in the global literature indicate that the efficiency, confidence, and sustainability can be greatly increased with the help of an online supply network and a traceability system [3], [4], [5].

It creates a compelling argument into the development of a digital Millet Supply Chain Platform that would combine the concept of the value chain and the implementation of the present supply chain technology. These holes, which are already reported, can be resolved directly through such a system: a more transparent market, a gap between the stakeholders, traceability, and fortifying the entire millet ecosystem.

III. METHODOLOGIES

The design of the proposed digital millet supply chain infrastructure has been developed in an organized system development methodology of practical feasibility, modular design, and iterative validation. The approach focused on the experiential study of practical agricultural issues and their conversion into practical system constraints and deployment of a working prototype to test usefulness and performance.

A. Requirement Analysis

The requirement analysis stage was meant to learn about the issues recorded in literature- broken market linkages, absence of transparency, frail traceability and the necessity to digitalize the system and convert them into system specifications. According to the needs of the stakeholders observed in the study of value chains [3], [10], the requirements were separated into three categories: The aim of the requirement analysis phase was to be familiar with the problems that were identified in literature; i.e. broken market connection, lack of transparency, poor traceability, and need of digital integration and transform them into real functionality of the system. They were divided into three groups according to the needs based on the requirements of all the stakeholders present in the value chain analysis [3], [10].

1. Functional Requirements

- Farmers, Customers and Admin Role, Startup Owners.
- Product listing modules are processed goods and by-products of ACM.
- Orders, integration with payment gateway (UPI/COD) and tracking of delivery.
- Checking- Profile and Aadhaar certifying by the government.
- Feedback and rating system.
- Recipe spreading materials and contact with the customers.
- Approval, analytics approval and analytics dashboard.

2. Non-Functional Requirements

- The excellent performance will be applicable on the rural consumers who have low rates of digital penetration.
- Encrypted and secured data storage and authorization.
- Scalability, and responsive page loading.
- None of the faults in flow of transactions.
- Constant information that has a strong background.

3. Stakeholder Requirements

- The farmers need clear pricing, confirmed buyers and easy tracking of orders.
- The owners of start-ups must have the supply of the raw materials and accessibility to the processed millet products.
- Customers seek genuine products, nutrition and provenience.
- Platform governance and verification processes must be handled fully by admins.

B. System Design Approach

The system design is based on a modular, layered architecture, which is consistent with the ideas of the literature on digital supply chain [4].

1. Layered Design

- Presentation Layer- User Interface/User-Experience.
- Application Layer- Business logic of orders and payments and verification.
- Data Layer- Secure data flow in centralised database.

2. Design Principles

- Ability to facilitate expansion in the registering of farmers and startups.
- Interoperability to interfere certification information and traceability data.
- Sustainability in parallel ACM models of waste-to-value.
- Centered design that can be acceptable to farmers with varying literacy level.

3. Workflow

Each of them (farmer, startup, customer, admin) is developed as a distinct functional component that is presented via a separate dashboard, yet linked to a central data structure that ensures a smooth interaction.

C. Development Process

The progression was done in a step-by-step workflow process that was useful in organizing the system in an efficient manner as well as giving a chance to make improvements at every stage.

1. Information Gathering and Requirement Study

This was initiated by gathering the extensive requirements based on the literature research, stakeholder consultations and on the response of our project reviewer. This assisted us in clarifying clearly what each user role, i.e. Farmer, Startup Owner, Customer and Admin, required of the system.

2. User Interface (UI) Design

After the requirements were clear, we proceeded on to the design of the user interface. All major modules such as product listing, dashboards, order pages, and admins had wireframes and screen layouts developed. The aim was to maintain the interface as simple, friendly to the mobile device, and easy to understand among the users with different degrees of digital literacy.

3. Backend Development and Integration

Backend was developed after the completion of the UI. The orders, payments, verification, and feedback business logic, the database structure, APIs and authentication modules were implemented. The system was then progressively linked to backend services by connecting each component of UI.

4. System Integration

Each of them was incorporated into one platform (farmer, startup, customer, admin, payments, and ACM). Data flow between modules was also tested to check out the consistency, ease of navigation and proper handling of transactions.

5. Testing and Refinement

There were several stages of testing which included: functional, usability, error handling and performance testing. Any identified problems were fixed, and the workflow could be streamlined to enhance the general usability particularly by farmers and those who were using digital equipment the first time.

D. Testing and Validation

During the development, testing was conducted to make sure that the platform was developed and functioning. Each module underwent basic functionality testing and the modules were then integrated together when the frontend and the backend have been linked. The usability was checked in order to ensure that the interface was easily navigated by farmers, buyers and administrators. Output validation was done through the examination of outputs, error correction, and feedback of project guide/reviewer.

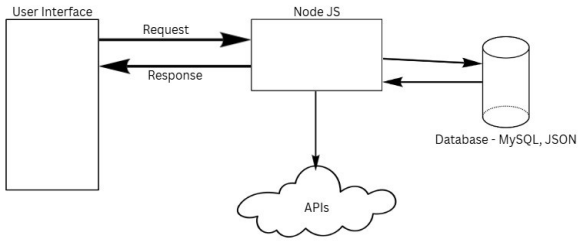
E. Tools and Environment

The site has been built on a newer stack of web technology:

- Frontend: HTML, CSS, JavaScript, React
- Backend: Node.js (Express)
- Database: MySQL
- External Services: Google Translate API
- Development Tools: VS Code, Git/GitHub
- Operating Environment: Windows 10/11

IV. System Design and Implementation

A. System Architecture



The architecture of the system consists of three layers in order to provide modularity, scalability and simplicity to maintain:

- Presentation Layer:
  The user interface was developed with the help of React and offers farmers, consumer, and administrator pages.
- Application Layer:
  The functionality implemented on the backend with Node.js/Express and that services authentication, product management and orders, payments and the generation of certificates.
- Data Layer:
  MySQL database with user, product, transaction and certificate as well as information related to millet.

This layer independent architecture enables all the layers to be modified or scaled without any impact on the other parts of the system.

B. Functional Modules

1. Farmer Module

- Farmer registration & login
- Placing the products, quantity and prices of the millet.
- The performance of the products, orders as well as payments.

2. Consumer Module

- Registration & login
- Traveling through millet products and nutritional/health benefits.
- Making payments and ordering.

3. Admin Module

- Consumer account management and Farmer.
- Acceptance/ check of product listing.
- Validating certificates
- Monitoring performance and dealings of the platform.

4. Payment Module

- Secure online buying on consumer orders.
- Farmers are advised of their payment statuses.
- Linkage to third-party/UPI-based gateways.

5. Certificate Module

- Promotes integrity and credibility of value chain.

6. Health Benefits Module

- Demonstrates scientifically-proven health benefits of all forms of millet.
- Helps the consumers to choose well in their diets.

7. Government Schemes Module

- Gives the recent developments on the plans of millet.
- Helps farmers to receive subsidies, training and incentives.

C. *Data Flow and Interaction*

The system data flow will be drawn to make the database, which is handled by the back end and front end interacts with each other without any difficulty:

1. User Interaction:
   The requests are provided using the React interface by users (farmer/consumer/admin).

2. Backend Processing:
   The request is dispatched to the Node.js/Express server in which the inputs are checked and necessary operations are performed (authentication, product management, order creation, etc.).

3. Database Operations:
   The backend is updated or queried against the MySQL database when the need arises.

4. Response Handling:
   The result of the processing is directed back to the frontend and displayed to the user.

This stream will ensure that data is updated in real time, interaction is safe, and system behavior is not problematic.

D. *Implementation Details*

- Frontend Implementation:
  Components in react were crafted to accommodate each user class with pure UI designs. Forms, product cards, and dashboards were made to have reusable components.
- Backend Implementation:
  The login, product management, payment, and certificate handling API were designed in the form of RESTful using Node.js/Express.
- Database Implementation:
  Users, products, orders, payments as well as certificates were organized into MySQL tables in appropriate relationships.
- Integration:
  The frontend and backend interact with the backend and MySQL respectively by using API calls and SQL queries respectively.
- Testing & Deployment:
  Testing was done on each module and subsequently integrated.

V. Results and Discussions

A. *Usability and User Experience*

The created platform was piloted with sample users who constituted the farmers, consumers, and admins. The interface was discovered to be easy to navigate through, intuitive and simple. Farmers could easily post the products, and the consumers could view and order millets by using few procedures. The layout was clear, responsive and the various modules were organized that made the user experience enjoyable.

B. *System Performance*

The system worked effectively in normal usage conditions. MySQL database retrieval was quick and API reasonably constant when interacted with a few times. The loading of pages was low and the backend was able to process user authentication, upload products as well as process orders without delays. The modular design was useful in maintaining similarity in performance of various portions of the platform.

C. *User Benefits*

The benefits it offers to the users have a number of direct advantages:

- Farmers will have an available online market to present their millet products and sell them.
- Consumers are provided with clear information regarding the quality, price and health benefits of products.
- Admins are able to track the activities efficiently, check listings and ensure reliability of the platform.

All in all, the system helps to increase the visibility, confidence, and convenience in all players in the millet value chain.

D. *Improvement over Existing System*

The proposed platform brings significant changes to the conventional millet marketing which, mostly, is offline, disorganized, and relies on middlemen:

- Farmer to consumer direct contact, less exploitation.
- Online traceability, enhancing quality and price transparency.
- Immediate availability of government plans and certifications that can be difficult to locate on-line.
- Information on health benefits, which allows consumers to make informed choices, is centralised.
- Quick transaction and communication, which is made possible by the online system.

Such improvements prove that the platform does not only digitise the value chain, but also makes it more efficient, reliable, and farmer-oriented.

VI. Social, Ethical and Sustainability Considerations

A. *Social Impact*

The social benefits of having the millet value chain platform are likely to be enormous. The system will address the issue of dependency on middlemen by allowing direct farmer entry to digital markets, which will contribute to the balanced income distribution. It empowers small and marginal farmers particularly rural and tribal areas through better market accessibility, visibility and bargaining power. Consumers also enjoy better transparency on origin of products, quality as well as price which results in generation of trust and informed decision making.

B. Ethical Considerations

The responsible information treatment of user data and open system practices would be considered an ethical way of implementing the platform. The data about the user (contact details, records of transactions and product data) should be stored and used with the legitimate purposes. The platform is not designed to be discriminatory and biased and offers equal access to all the groups of farmers without regards to their regions, gender, or the magnitude of production. Good communication on certifications, quality of the products, and pricing is ensured to avoid miscommunication or unethical marketing.

C. *Sustainability Implications*

Some of the known ecological benefits of millets include, low water needs, resistance to harsh climatic conditions, and the ability to fit well in sustainable farm systems. The platform indirectly leads to environmental sustainability by encouraging people to consume millet and making markets linked effectively. The electronic system also saves the necessity of physical intermediaries, inefficiency in transportation, and paperwork. Promotion of value-added millet products and promotion of sustainable practices of farmers also resonates with other Sustainable Development Goals (SDGs), especially SDG 2 (Zero Hunger), SDG 12 (Responsible Consumption and Production), and SDG 8 (Decent Work and Economic Growth).

VII. Conclusion

The suggested millet value chain platform offers a systematic and technology-oriented solution to the predicaments that have endured in the millet market. The system allows farmers, consumers, and administrators to be connected to one digital system which increases transparency, enhances accessibility and lowers dependency on intermediaries. The modular system, as well as convenient interface, make sure the basic features of the system, including product listing, payments, certification display and access to government schemes, are effectively delivered.

The findings show that the platform is not only efficient in facilitating transactions but also encourages informed decision-making as products are well traced and their information made clear. This would mean increased market visibility and fair prices to the farmers and trust and convenience to consumers. In a wider sense, the system helps the social equity, ethical use of data and sustainability by promoting the production of millet that is resistant to climate changes and low-impact consumption.

All in all, the project proves that the digital intervention may become a huge boost to the millet value chain and a viable template that can be used to build upon in the future, such as the integration with mobile, sophisticated analytics, and additional services to support farmers.

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